Trent Haines

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Experience

Roblox — **Feb 2024 – Present**

Software Engineer - Backend, Payment Systems — *San Mateo, California*

- Delivered a zero-downtime migration of 10B+ core billing entity records. Led end-to-end data validation, created interface adapters, and refactored legacy systems to maintain seamless compatibility
- Built a drop-in dual-read/dual-write framework adopted as the organization standard across 10 engineering teams. Abstracted away cutover complexity through configurable execution modes, background orchestration, error handling, rollout controls, mismatch telemetry, and automatic alerting
- Evolved subscription renewal management from an unreliable table-scan processor to a workflow-driven Temporal orchestrator, improving availability from <99% to >99.99%
- Traced the full service communication network for Payment Search across 3B+ payment records and multiple payment integrations, revealing redundant paths and fragmented data sources. Rearchitected Payment Search to directly communicate with a new Elasticsearch-backed API, eliminating middlemen and establishing a single source of truth
- Redesigned the internal representation of 'product', replacing an inflexible 6-table relational schema with a unified single-table with Protobuf-structured data, serving as the data layer for all real-money transactions on Roblox platform

Amazon — **May 2022 – August 2022**

Software Engineering Intern - Fullstack, Route Planning Optimization — *Austin, Texas*

- Developed an interactive web app enabling Amazon science teams to analyze delivery route execution patterns, guiding data-driven route-planning optimization decisions
- Identified 14 operational metrics (ex: aggregate package weight, delivery density, average stop duration) and implemented end-to-end data pipelines from database queries to front-end visualizations supporting flexible route comparison

FireEye — **May 2021 – August 2021**

Software Engineering Intern - Backend, Telemetry — *Dallas, Texas*

- Modernized the container infrastructure for a core threat detection engine by executing a full migration from ECS to EKS, improving operational control and scalability
- Built comprehensive monitoring for the newly deployed EKS-based engine featuring Grafana dashboards and PromQL metrics to track container performance, throttling events, and resource utilization

University of Texas at Dallas — **May 2016 – August 2017**

NanoTechnology Research Intern — *Richardson, Texas*

- Developed applications for thermally actuated nylon artificial muscles, with prototypes incorporated into a successful NASA STTR Phase II grant application
- Synthesized organic aerogels and conducted experiments to test and optimize their material properties

Education

University of Texas at Dallas — **August 2019 - May 2023**

Master's + Bachelor's of Science in Computer Science — *Richardson, Texas*

- 4-Year Fast-Track MS/BS
- Full Ride Merit Scholarship
- Computing Scholar's Honors

Relevant Coursework

- Algorithm Analysis
- Distributed Computing
- C++ in UNIX Environment
- Statistical Methods for AI/ML
- Database Systems
- Operating Systems

Projects

Convex Hull Algorithm Visualizer | *JavaScript, HTML, CSS* — **April 2022**

- Created a React application that visualizes three different Convex Hull algorithms to help Computational Geometry students understand and compare them through custom data-sets that highlight each algorithm's strengths in various scenarios

Blind Accessibility Chrome Extension | *JavaScript, Python* — **April 2020**

- Collaborated with a team to create a Chrome extension using Google's Cloud Vision API to provide text labeling for images on webpage. These labels can be read aloud using text-to-speech, or can assist in image captioning

Technical Skills

Languages: C#, C++, Python, Java/Kotlin, JavaScript/TSX
Frameworks and Libraries: ASP.NET Core, gRPC/Protobuf, Temporal Workflows
Tools: Git, Linux, Docker, Grafana